

Assignment 5

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Set Up

Packages

```
#general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
library(scales)
library(NatParksPalettes)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

Necessary Code from Class

```

# creating new object from sites
ncos_sites <- sites %>%
  # filtering to only include sites at NCOS, sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
    case = "snake"))

# creating a new object from water quality
water_quality_clean <- water_quality %>%
  # cleaning column names
  clean_names() %>%

```

```

# filtering join: only include sites in ncos_sites
semi_join(ncos_sites, by = "site") %>%
# creating new column to join with later data frame
unite("date_site", date, site, remove = TRUE) %>%
# grouping by date/site
group_by(date_site) %>%
# calculating median pH, salinity, and DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
# ungrouping the data frame
ungroup() %>%
# filtering out salinity outliers
filter(date_site != "2022-08-12NVBR")

# creating a new object from aq_ins
aq_ins_clean <- aq_ins %>%
# cleaning column names
clean_names() %>%
# filtering join: only include quarterly NCOS surveys from ncos_site
semi_join(ncos_sites, by = "site") %>%
# renaming column
rename(date = date_on_vial) %>%
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       annelida_polychaete,
       diptera_chironomid) %>%
# filtering sample types
filter(sample_type %in% c("FB250", "FB 250")) %>%
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
)

```

```

) %>%
group_by(date, site, taxon) %>%
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
# ungrouping the data frame
ungroup() %>%
# created new column to join with data frame
unite("date_site", date, site, remove = FALSE) %>%
# joined with water quality clean
left_join(water_quality_clean, by = "date_site") %>%
# joined with taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
# making a new column called site_full with proper names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NDC" ~ "Devereux Creek")) %>%
# reordering site full by median salinity
mutate(site_full = fct_reorder(site_full,
                              med_sal,
                              .fun = "median",
                              .na_rm = TRUE),
       # reversing the order of site_full
       site_full = fct_rev(site_full))

# creating new object from aq_ins_clean
copepods <- aq_ins_clean %>%
# filtering to only include copepod observations
filter(taxon == "copepod") %>%
# creating a new column that takes the natural log of average abundance per
↪ liter
mutate(log_ave_lit = log(ave_lit + 1))

```

1. Visualizing differences across sites in major taxon

a. Planning the Figure

- x-axis: Salinity (ppt)

- y-axis: taxon
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary()

b. Wrangling the data

```
# creating a new object called ostracods
ostracods <- aq_ins_clean %>%
  # filtering to only include ostracods
  filter(taxon == "ostracod") %>%
  # creating a new column that takes the natural log of average abundance per
  # liter
  mutate(log_ave_lit = log(ave_lit + 1))

# taking a random sample from the new data frame
slice_sample(ostracods, n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	ostr~	0.133	7.71	1.98	9.59
2	2023-05-04_NEC	2023-05-04 00:00:00	NEC	ostr~	0.667	8.05	NA	8.55
3	2023-09-15_NMC	2023-09-15 00:00:00	NMC	ostr~	0.4	NA	NA	14.2
4	2023-08-12_NEC	2023-08-12 00:00:00	NEC	ostr~	0	NA	NA	NA
5	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	ostr~	0	8.86	45.2	10.2
6	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
7	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
8	2021-04-15_NPB	2021-04-15 00:00:00	NPB	ostr~	185.	8.09	3.57	8.24
9	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.267	NA	NA	NA
10	2022-02-25_NEC	2022-02-25 00:00:00	NEC	ostr~	0	8.75	44.1	9.87

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

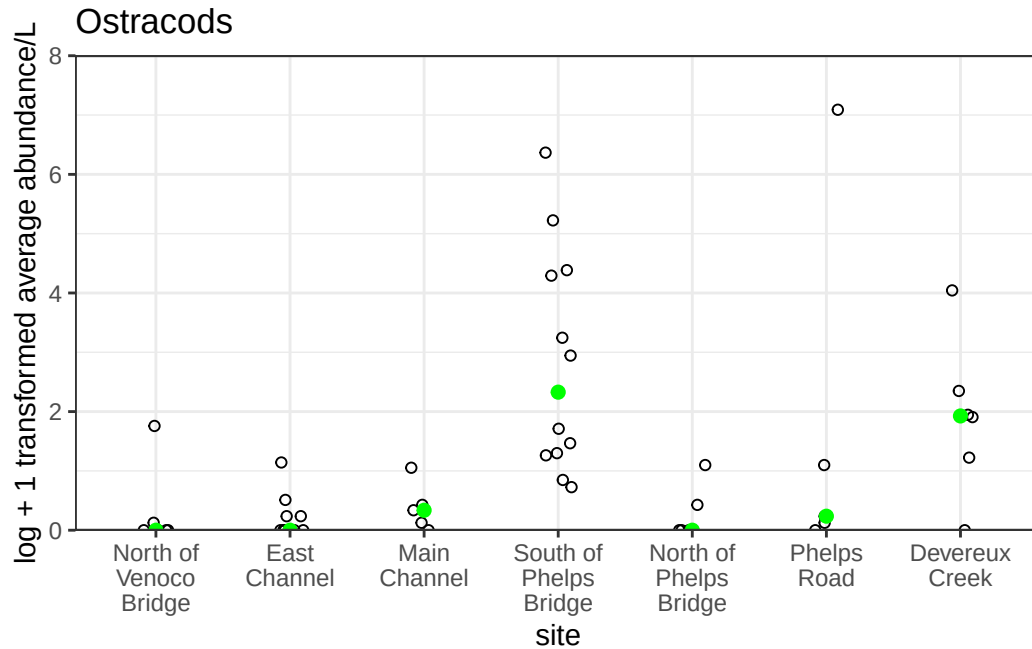
c. Coding the figure

```

# creating a new object that stores the ostracod ggplot
ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit))+
# plotting ostracod observations with geom_jitter
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# adding a distinct green median point
stat_summary(geom = "point",
            fun = median,
            color = "green",
            size = 2)+
# getting rid of overlapping labels
scale_x_discrete(labels = label_wrap(width = 10)) +
# getting rid of y-axis space at the bottom of the graph
scale_y_continuous(expand = c(0, 0),
                  limits = c(0, 8)) +
# adding labels
labs(x = "site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# getting off the default theme
theme_bw()

ostracod_plot

```



d. Creating a multipanel figure

```
# creating an object of a salinity plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
  # plotting salinity measurements with geom_jitter
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # adding a distinct red median point
  stat_summary(geom = "point",
              fun = median,
              color = 'red') +
  # getting rid of overlapping labels
  scale_x_discrete(labels = label_wrap(width = 10)) +
  # getting rid of y-axis space at the bottom
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 200)) +
  # adding labels
  labs(x = "site",
```

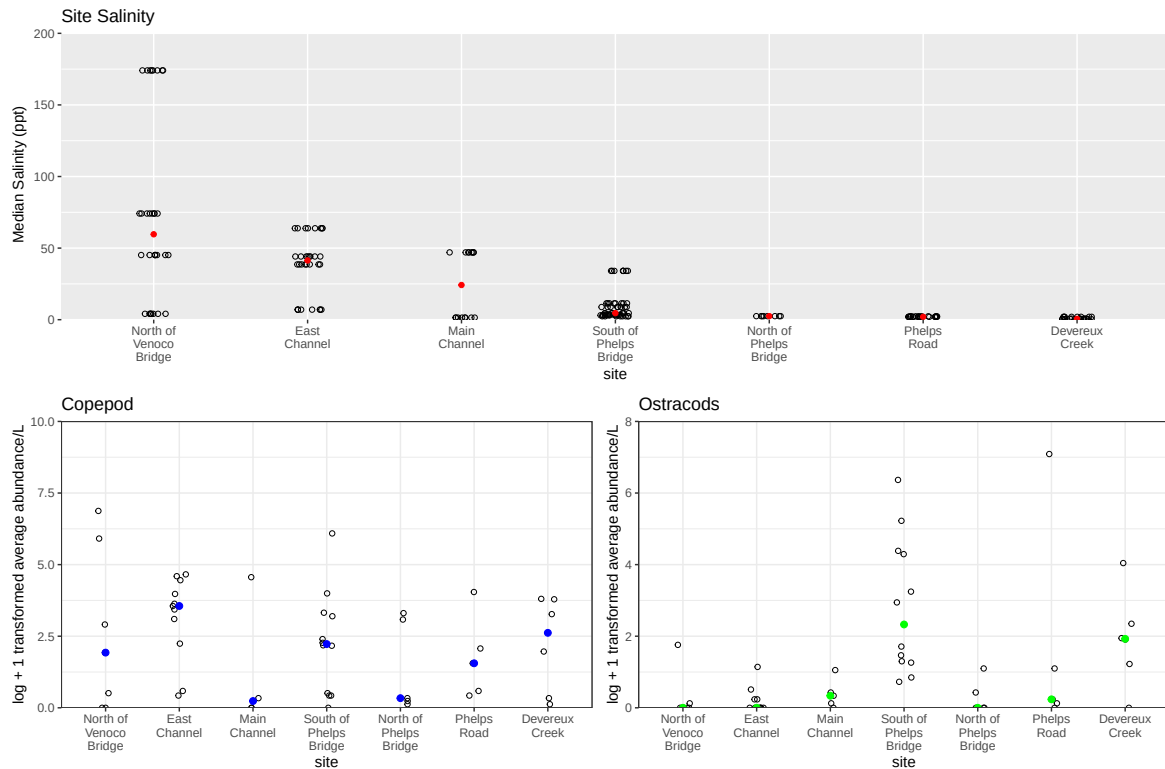
```

    y = " Median Salinity (ppt)",
    title = "Site Salinity")

# creating a plotting object of copepod abundance
copepod_plot <- ggplot(data = copepods,
                      mapping = aes(x = site_full,
                                    y = log_ave_lit))+
# plotting copepod observations using geom_jitter
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# adding a distinct blue median point
stat_summary(geom = "point",
            fun = median,
            color = "blue",
            size = 2)+
# getting rid of overlapping labels
scale_x_discrete(labels = label_wrap(width = 10)) +
# getting rid of y-axis space at the bottom
scale_y_continuous(expand = c(0, 0),
                  limits = c(0, 10)) +
# adding labels
labs(x = "site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepod") +
# getting off the default theme
theme_bw()

# printing a multi panel view of my 3 plots
salinity_plot /
(copepod_plot | ostracod_plot)

```

e. Interpreting the figure

The biggest differences in copepod and ostracod abundance across sites can be seen at the East Channel site, where copepods are highest in abundance, and ostracods are extremely low in abundance. The two species are similar in abundance at the Main Channel, South of Phelps Bridge, North of Phelps Bridge, and Devereux Creek sites.

There doesn't appear to be a correlation between site salinity and copepod or ostracod abundance. Median salinity is order by descending ppt from left to right on the x-axis, and the invertebrate abundance plots with the same x-axis don't follow a descending or ascending pattern, appearing relatively random.

2. Visualizing total abundance as a function of salinity

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: log_avg_lit

- color: taxon
- geometry: geom_points()
- facets: facet_wrap(~ taxon)

b. Wrangling the Data

```
# creating a new object from aq_ins_clean
avg_taxon_abundance <- aq_ins_clean %>%
  # filtering out na salinity rows
  filter(!is.na(med_sal)) %>%
  # filtering out rows with 0 invertebrate observations, better for
  ↪ visualization
  filter(!ave_lit == 0) %>%
  # selecting relevant columns
  select(date_site, taxon, med_sal, ave_lit) %>%
  # creating new columns, correcting taxon column
  mutate(
    # converting taxon to title case
    taxon = to_any_case(taxon, case = "title"),
    # taking natural log of average abundance per liter
    log_ave_lit = log(ave_lit + 1),
    # reorder taxon column based on max average abundance
    taxon = fct_reorder(taxon, ave_lit, .fun = "max", .na_rm = TRUE),
    # reversing order so that top rows appear first
    taxon = fct_rev(taxon))

# printing a random sample of 10 rows
slice_sample(avg_taxon_abundance, n = 10)
```

```
# A tibble: 10 x 5
```

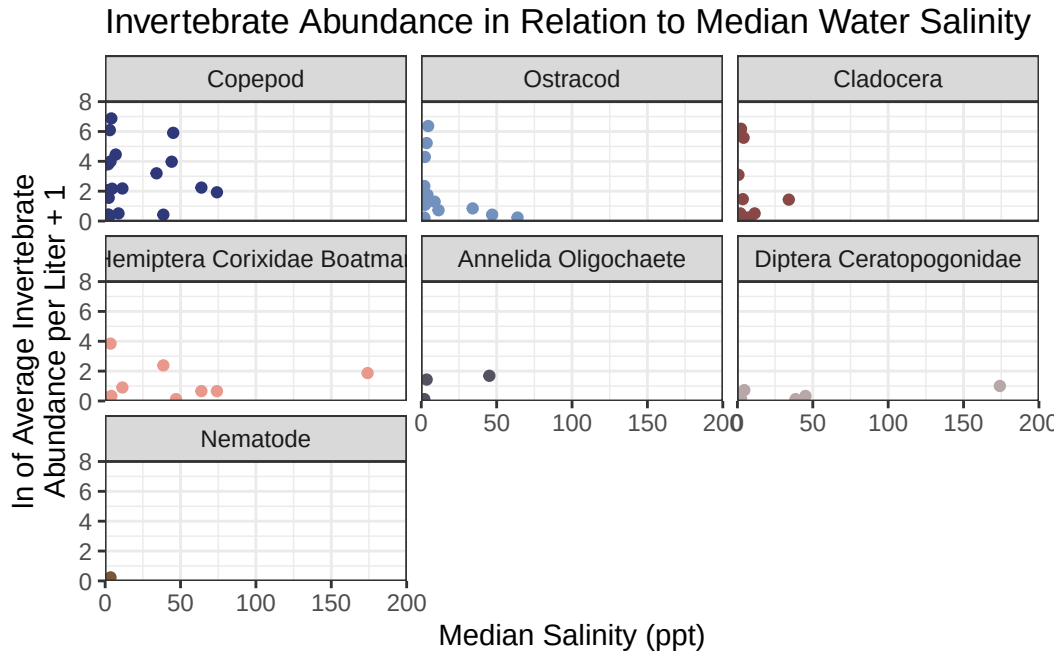
	date_site	taxon	med_sal	ave_lit	log_ave_lit
	<chr>	<fct>	<dbl>	<dbl>	<dbl>
1	2023-03-03_NPB2	Copepod	1.98	0.533	0.427
2	2020-01-21_NDC	Copepod	0.3	43.2	3.79
3	2021-11-23_NPB2	Cladocera	1.99	0.667	0.511
4	2023-03-03_NDC	Copepod	0.688	6.13	1.96
5	2022-11-12_NEC	Copepod	7.01	85.1	4.46
6	2022-02-21_NPB	Copepod	34.1	23.5	3.20
7	2022-11-12_NEC	Cladocera	7.01	0.267	0.236
8	2020-02-21_NVBR	Hemiptera Corixidae Boatman	4.1	0.4	0.336

9	2022-02-21_NPB	Cladocera	34.1	3.2	1.44
10	2024-01-28_NPB	Ostracod	3.08	3.33	1.47

c. Coding the figure

```
# plotting avg_taxon_abundance with salinity on the x-axis and average
  ↳ abundance on the y-axis
ggplot(data = avg_taxon_abundance,
        mapping = aes(x = med_sal,
                       y = log_ave_lit,
                       color = taxon))+

# plotting
geom_point()+
# making a new panel for each taxon
facet_wrap(~ taxon)+
# getting off the default theme
theme_bw()+
# adding custom color palette
scale_color_manual(values = natparks.pals("Torres"))+
# removing the legend
theme(legend.position = "none")+
# removing space on the y-axis at the bottom
scale_y_continuous(expand = c(0,0),
                   limits = c(0,8))+
# removing space on the x-axis
scale_x_continuous(expand = c(0,0),
                   limits = c(0,200))+
# adding labels
labs(x = "Median Salinity (ppt)",
     y = "ln of Average Invertebrate\nAbundance per Liter + 1",
     title = "Invertebrate Abundance in Relation to Median Water Salinity")
```



d. Interpreting the Figure

All taxa appear to prefer water with salinity between 0 and 100 ppt, as most data points are clustered within that range. The invertebrates that are most abundant at relatively low salinity levels (0-50 ppt) are Copepods, Ostracods, and Cladocera, who all spike in abundance, with Copepods nearly reaching a maximum of 966 individuals per liter at only 4.1 median salinity. There are two notable exceptions to the trend of high abundance in low salinity - Hemiptera Corixidae Boatman and Diptera Ceratopogonidae were both observed at fairly high abundance on August 8th 2022 when median salinity was at 174 ppt, while all other species were not observed. Overall, this figure shows that invertebrates are found in higher levels of abundance when salinity is low.